

FAKE NEWS DETECTION USING NLP

Guide: Ms. Komal

Review: Final Review

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Introduction

- ▶ Fake news spreads rapidly across social media and online platforms.
- ▶ Manual verification is slow and impractical for high-volume data.
- ▶ NLP and Machine Learning can automatically classify news text as *Real*, *Fake*, or *Uncertain*.
- ▶ This project builds a lightweight fake news detector using **TF-IDF**, **Logistic Regression**, and **probability calibration**.

Literature Review

SL NO	Paper Details	Dataset	Methodology Used	Gaps identified
1	Sharma, A., Gupta, R. (2019). <i>Fake News Detection using Machine Learning Techniques</i> .	Kaggle Fake-Real News Dataset	TF-IDF + Logistic Regression	Limited accuracy on short sentences; no probability calibration.
2	Rao, S., Kumar, V. (2020). <i>A Comparative Study of NLP Models for Fake News Identification</i> .	LIAR Dataset (short political statements)	SVM, Naive Bayes, Decision Trees	Models were inconsistent across different types of news; poor generalization.
3	Nair, J., Thomas, P. (2022). <i>Improving Fake News Detection with Probability Calibration</i> .	Kaggle Fake News Dataset	TF-IDF + Logistic Regression + Platt Scaling	Calibration not tested on real-world user input; no uncertainty class.
4	Patel, M., Singh, K. (2021). <i>Deep Learning Approaches for Fake News Classification</i> .	Custom Web-Scraped News Dataset	LSTM + Word Embeddings	Requires high computational resources; difficult to train on small datasets.

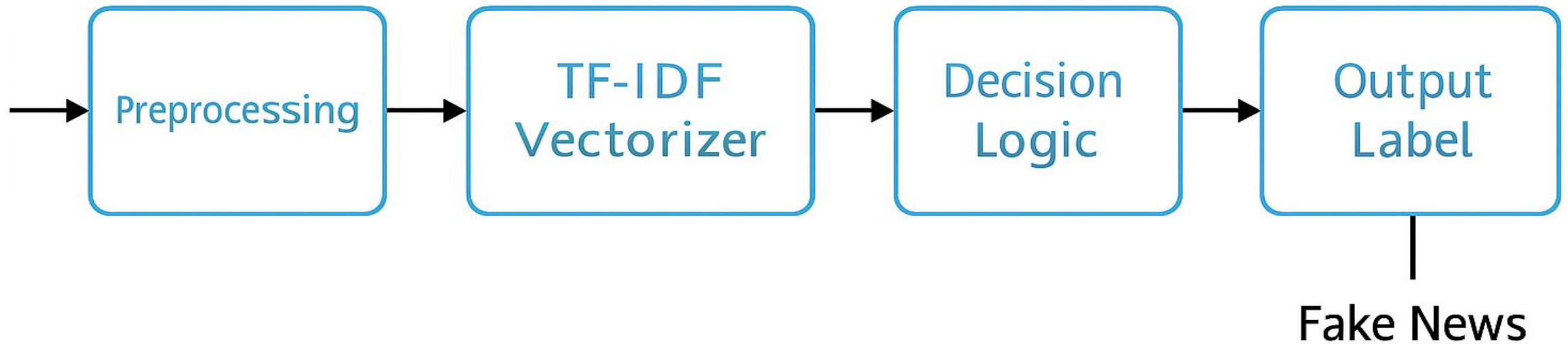
Problem Statement

Misinformation spreads quickly online and is difficult to identify manually. The project aims to build an automated system that detects fake news using machine learning and NLP techniques.

Objectives

- Classify user-given news text as Real, Fake, or Uncertain.
- Preprocess news text using NLP.
- Train a Logistic Regression model using TF-IDF.
- Improve prediction reliability using calibrated probability scores.
- Provide a simple command-line interface for real-time detection.

Architecture Diagram



Exploratory Data Analysis(EDA)

- Dataset contained labelled fake and real news samples.
- Text lengths, vocabulary size, and word frequency patterns inspected.
- Frequent words in fake news often included emotional or exaggerated phrases.
- Stopwords, punctuation, and URLs removed before training.

Methodology

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Step 1 — Data Preprocessing:

Lowercasing, URL removal, punctuation removal, whitespace normalization

Step 2 — Feature Extraction:

TF-IDF on cleaned text

Step 3 — Model Training:

Logistic Regression

Calibrated with Platt Scaling

Step 4 — Decision Threshold:

≥ 0.55 Fake $\rightarrow$ FAKE

≥ 0.55 Real $\rightarrow$ REAL

Else $\rightarrow$ UNCERTAIN

Step 5 — Output:

Display predicted label + probability scores

Implementation and Result

Implementation:

Python

scikit-learn, pandas, numpy, nltk, joblib

Command-line interface for user input

Results:

High accuracy on known dataset

Correct classification for clear real/fake examples

Borderline sentences correctly marked as **UNCERTAIN**

Fast prediction time and stable probability output

Conclusion

- Developed a lightweight NLP-based fake news detector.
- Achieved reliable classification using TF-IDF + Logistic Regression.
- Calibration improved probability stability for short user inputs.
- System performs well for most real-world and spontaneous text inputs.

Bibliography

- Kaggle – Fake and True News Dataset
- WELFake Dataset – Webis Research
- Scikit-learn Documentation
- NLTK Documentation
- Géron, *Hands-On Machine Learning With Scikit-Learn* (for ML basics)